Digital Image Processing and Analysis

Overview: Digital Image Processing

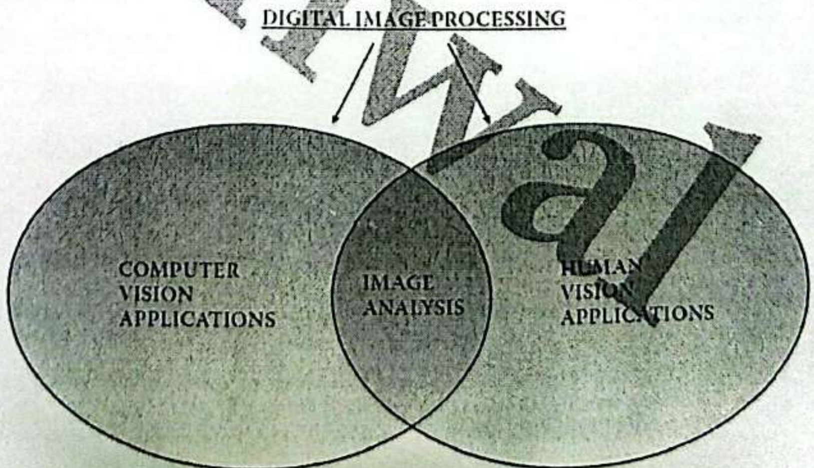
- Another *important aspect* of ***computer imaging*** involves the *ultimate "receiver"* of the visual information
  - in some cases the human visual system,
  - in others the computer itself.
- This distinction allows us to separate digital image processing into two primary application areas:
  - (1) **human vision applications**, and
  - (2) **computer vision applications**,

Overview: Digital Image Processing

- Definition of *Digital image processing,* also referred to as **computer imaging**:
  - Acquisition and processing of visual information by computer.
- Why is it important?
  - Human primary sense is visual sense.
  - Information can be conveyed well through images (one picture worth a thousand words).
- Computer is required because the amount of data to be processed is huge.

Overview: computer imaging

- *Image analysis* is a key component in the *development of both*



Overview: computer imaging

- In *computer vision* applications the processed (output) images are for use by a computer, while
- In *human vision* applications the output images are for human consumption.

Computer Vision

- *Image segmentation* is often one of the first steps in finding higher-level objects from the raw image data.
- *Feature extraction*: acquiring higher-level image information (shape and color)
- *Pattern classification*: using higher-level image information to identify objects within the image.

1.2 Image Analysis and Computer Vision

- Image analysis involves the *investigation of the image data for a specific application.*
- The *image analysis* process for computer vision requires the *use of tools* such as
  - *image segmentation*,
  - *image transforms*,
  - *feature extraction*, and
  - *pattern classification.*

Computer Vision

- Most computer vision applications involve tasks that:
  - Are *tedious* for people to perform.
  - Require work in a *hostile environment.*
  - Require *a high processing rate.*
  - Require access and use of a *large database of information.*

Computer Vision

- Examples of applications of computer vision:
  - Quality control (inspect circuit board).
  - Facial recognition..
  - Automatic counting and grading of lumber
  - Automatic diagnosis of skin lesions.
  - Automatic identification of fingerprints.
  - DNA analysis
  - And many, many others.

Examples

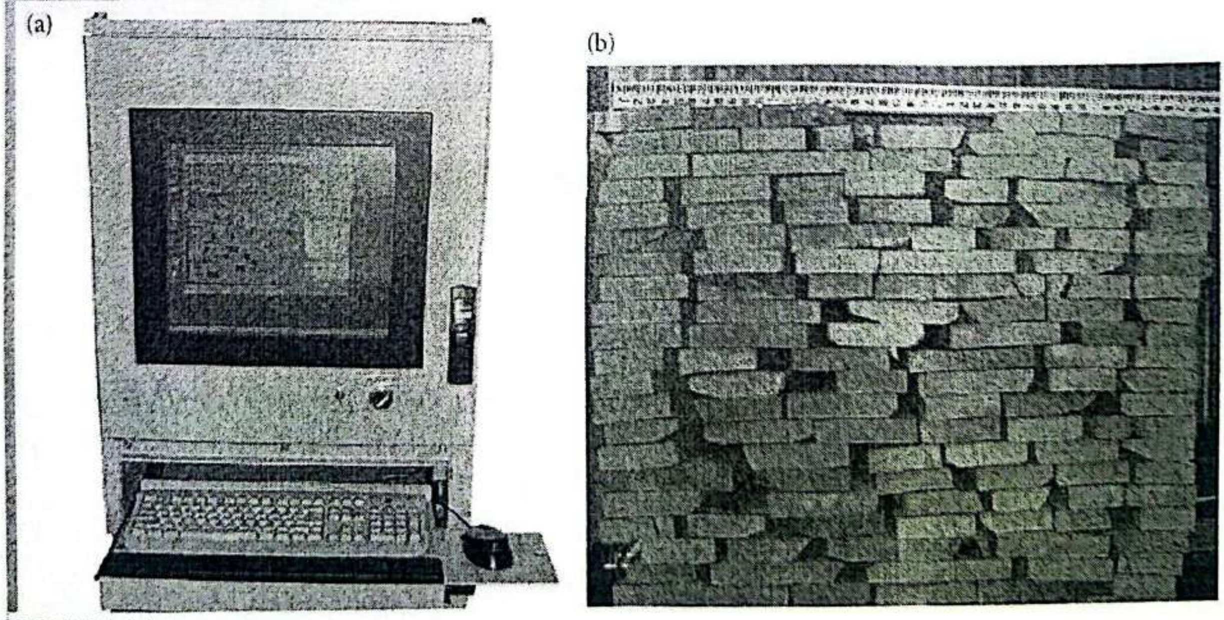
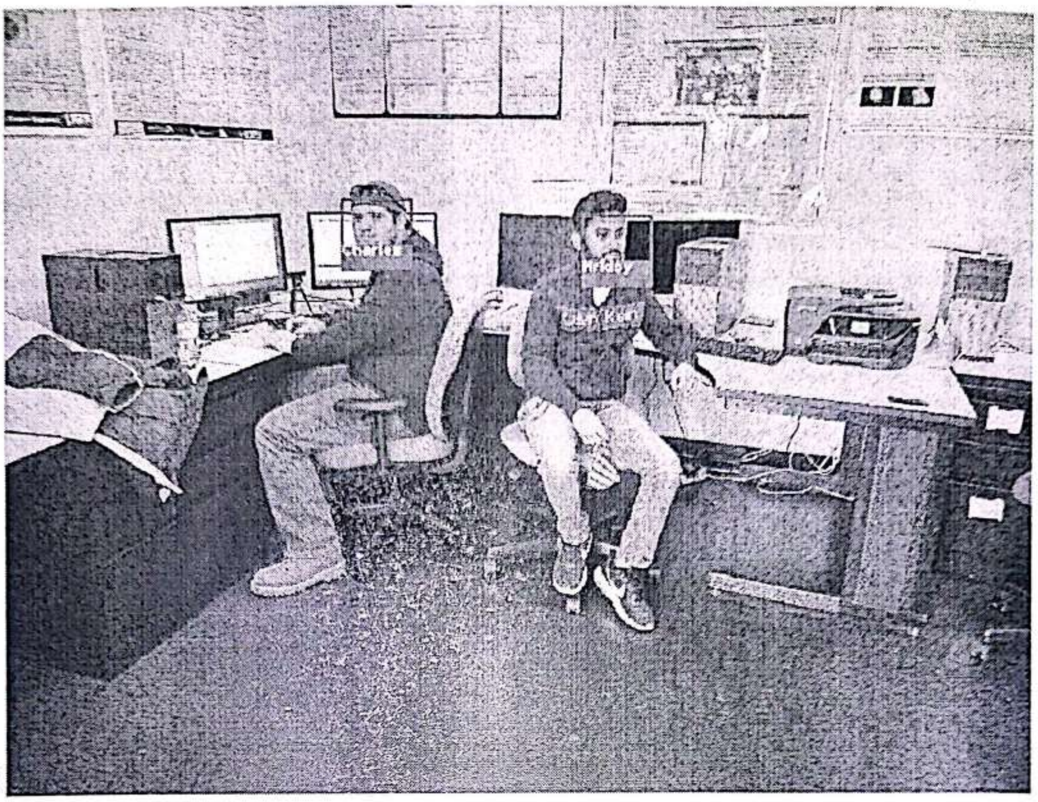
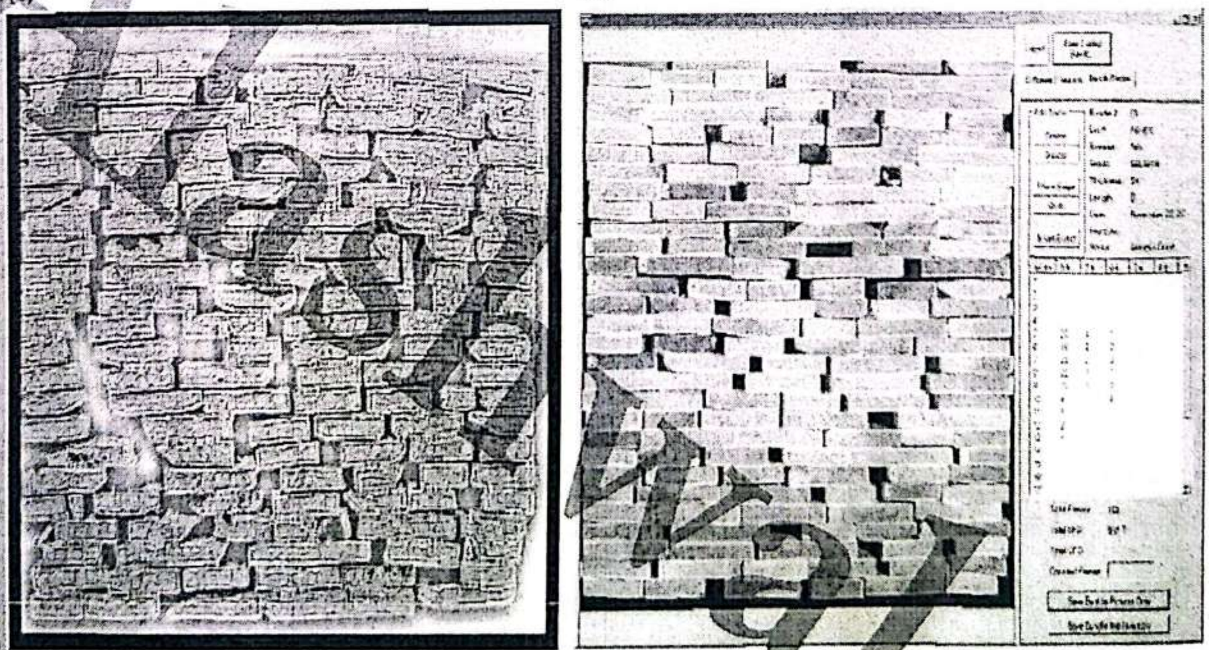


FIGURE 1.2-2
(a) Computer vision system for lumber counting and grading, (b) image captured by the system, (c) intermediate image after processing, (d) example of system output after software analysis. (Photos courtesy of Tony Berke, River City Software.)



(a)

(c) (d)



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